

Chaya Chandana Doddagalluru Appajigowda

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience specializing in Large Language Models (LLMs), agentic AI systems, and production ML pipelines. Designed, built, and deployed end-to-end AI solutions including LLM-powered health intelligence platforms, autonomous incident response agents, RAG-powered applications, and deep learning forecasting systems. Adept at Python rapid prototyping using PyTorch, LangChain, LangGraph and cloud platforms; experienced in taking models from research to production across NLP, time-series, and anomaly detection domains. Proven ability to architect scalable AI pipelines, collaborate with cross-functional engineering teams, and deliver measurable business impact.

WORK EXPERIENCE

AI Engineer - MyEdMaster	Feb 2026 – Present
- Designed and productionized AI-driven health intelligence systems, applying feature engineering, statistical modeling, and validation on 100+ clinical variables to deliver risk predictions and personalized recommendations, with model and data versioning for reproducibility. - Optimized end-to-end ML inference workflows by enhancing data pipelines, API orchestration, and execution efficiency, reducing response latency by 35% and enabling scalable, real-time delivery of personalized health insights. - Engineered LLM-powered RAG systems leveraging semantic search, vector embeddings, hybrid retrieval, and prompt engineering to enable context-aware AI interactions across personalized health workflows.	
ML Engineer (Student Technical Assistant) – GWU Law School	Jul 2024 – Dec 2025
- Built predictive models to forecast classroom recording demand and classification models to flag high-risk scheduling conflicts reducing last-minute recording failures by 25% and cutting manual follow-up work by 30%. - Developed anomaly detection pipelines to catch unusual recording access patterns and deployed model endpoints for real-time scheduling decisions, with experiment tracking across linear, tree-based, and neural network models.	
Machine Learning Engineer – Tata Consultancy Services	Nov 2021 – Dec 2023
- Built an anomaly detection system to monitor over 1M+ daily transactions, catching abnormal patterns early and reducing system downtime by 30%, fine-tuned detection thresholds using precision-recall analysis to cut false positives by 25%. - Engineered features like rolling transaction counts, frequency spikes, and deviation from historical baselines to make anomaly signals clearer, then trained and evaluated Isolation Forest and gradient-boosted models across time-aware validation splits. - Partnered with backend engineers to deploy models to production with proper logging, monitoring, and experiment tracking ensuring consistent results across retraining runs.	

TECHNICAL SKILLS

Programming & AI/ML Development: Python, ReactJS, Node.js, PyTorch, PySpark, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, LangSmith, NLP, LLMs, RAG, Generative AI, Agentic AI, Prompt Engineering.

Infrastructure & Distributed Systems: Microservice Architecture, Distributed System Design, Service Oriented Architecture, REST APIs, SRE Principles, DevOps Automation, Docker, Redis, Kubernetes, Containerization, Model Serving (FastAPI, Flask, Streamlit, Uvicorn).

Data & Cloud Platforms: AWS, Azure, GCP, Databricks, Snowflake, Apache Spark, Spark SQL, SQL, NoSQL (MongoDB, MySQL, Elasticsearch), Vector Databases (Pinecone), ELT Pipelines, Vector Databases (ChromaDB, FAISS), Data Warehousing, Data Lakehouse, Neo4j.

EDUCATION

The George Washington University Master of Science, Data Science	Washington, DC Dec 2025
Dayananda Sagar University Bachelor of Computer Applications	Bengaluru, India Sep 2021

PROJECT EXPERIENCE

Incident Response & On-Call Agent
Architected an Autonomous Incident Response & On-Call Agent leveraging **LangGraph** multi-node StateGraph orchestration, **Ollama (Mistral)** local **LLM**, and confidence-scored agentic reasoning loops that autonomously query logs and Prometheus metrics, retrieve remediation playbooks via **ChromaDB RAG**, execute kubectl auto-remediation, and store incident history in persistent vector memory deployed as a **FastAPI** REST service with Redis event-driven pub/sub ingestion, **LangSmith** full-trace observability, and **Docker** Compose single-command deployment, and automated **PagerDuty**, **Slack**, and **Jira** incident ticketing, reducing simulated MTTR from 45 minutes to under 5 minutes.

RAG-Powered Resume Intelligence System
Built a production-grade **RAG**-powered Resume Intelligence System using **LangChain**, **FAISS**, **HuggingFace** embeddings (**all-MiniLM-L6-v2**), and **Groq LLaMA 3.3 70B** engineered a per-file chunk grouping retrieval strategy to fix systematic document bias in multi-resume corpora, achieving **90%** answer faithfulness across 10 structured evaluation cases with keyword-match scoring logged to JSON.

Designed and deployed a multi-document Q&A API with a real-time chat UI via **Flask REST API** implemented custom prompt engineering, context-aware retrieval pipelines, and iterative chunking optimization (500→300 chars, overlap 50→100) that improved retrieval accuracy; deployed on **Railway** with dynamic port binding and version-controlled **FAISS** vector index, accessible via live production URL.

Forecasting School Meal Production Costs (IEEE Research Paper - Under Review)
Developed a comparative **ML/DL** forecasting system for Fairfax County Public Schools using **LSTM**, **GRU**, **XGBoost**, **FNN**, and Linear Regression across **177K+** records from **100+** schools - multivariate LSTM achieved best **RMSE** of 137 ($R^2 \approx 0.86$) using served, planned, discarded, and leftover meal features as inputs.

CERTIFICATIONS

Databricks Certified Data Engineer Associate | Databricks